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THE PALAEOMAGNETISM AND GEOLOGY OF THE PATREKSFJÖRDUR—ARNARFJÖRDUR REGION OF NORTHWEST ICELAND

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ABSTRACT

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The headlands between the fjords Arnarfjördur and Patreksfjördur in northwest Iceland, consisting mostly of Upper Tertiary plateau basalt lavas, have been geologically mapped in detail. Magnetic properties of samples from some 50 lava flows and dykes in the region have been measured; the mean palaeomagnetic pole position obtained from forty normally magnetized lavas is at 88°N113°E. The application of the present work to stratigraphic correlation and magnetic anomaly interpretation in northwest Iceland is discussed.

INTRODUCTION

The Tertiary basalt plateau of Vestfirdir, the northwest peninsula of Iceland, has received relatively little attention from earth scientists. Kjartansson's map (1969) summarizes his own and older geological observations; Hald et al. (1971) mapped a central volcanic complex at Króksfjördur, and Friedrich (1966) studied the lavas near Brjánslaekur, with special reference to the intercalated plant beds. Kristjánsson (1968) has studied the palaeomagnetism of lava flows near Isafjördur, and Moorbath et al. (1968) have dated samples from somewhat higher in the lava pile at 16.0 m.y. S. Jakobsson (quoted by Pálmason and Saemundsson, 1974) has obtained a date of about 13 m.y. in the region to be described below. Serson et al. (1968) have published high-altitude aeromagnetic profiles passing over the peninsula.

The present work is the first detailed mapping in northwest Iceland comparable in aim to the work of Walker and collaborators (Dagley et al., 1967) in east Iceland. Geological mapping in the Patreksfjördur—Arnarfjördur region

was carried out by J. Preston during the summers 1970—1973. The geology of some 600 sq. km of well dissected basalt plateau is illustrated in Fig. 1 and in the stratigraphic section of Fig. 2. L. Kristjánsson collected oriented core samples from 44 lava flows and 7 dykes in a continuous sequence near the middle of the succession, and measured their magnetic properties in collaboration with R. Pätzold.

GEOLOGY

Friedrich (1966) recognized four groups of plateau basalts in the Brjánslaekur area. These are labelled in alphabetic order and the dominant polarity is indicated by a suffix n or r.

	Description	Dominant polarity	Thickness
Top	feldsparphyric group Dn	normal	ca. 200 m exposed
	olivine basalt group Cr	reversed	100 m
	tholeiitic group Bn	normal	200 m
Bottom	feldsparphyric group Ar	reversed	>300 m

The lowest group (Ar) extends, up-dip, into the Patreksfjördur—Arnarfjördur region; it underlies the tholeiitic group of basalts on Skardabrún above Ósafjördur, which in turn is probably equivalent to Friedrich's Bn group. The base of the lithological group Ar is not exposed at the western limits of the region. R.M. Macintyre of the Scottish Universities Research and Reactor Centre has determined the K/Ar ages of three samples. His results are 11.9 ± 0.3 m.y. for the top flows of group Ar and 9.1 ± 0.3 m.y. for the lowest flows of group Bn.

Fig. 2 illustrates a generalized stratigraphy. Non-porphyritic and feldsparphyric basalts, with all gradations between, are the commonest lava type. The flows are 2—30 m thick, often with well oxidized, cindery tops. Average thickness of 400 flows measured is 11 m. Bytownite to labradorite plagioclase occurs as both phenocrystal and groundmass phase. Olivine appears sparsely as phenocrysts, and very rarely occurs in the groundmass. Chemically the suite plots as quartz normative basalts and follows a tholeiitic trend on the $FeO + Fe_2O_3/MgO + FeO + Fe_2O_3$—$SiO_2$ diagram. Flows of a distinctive and mappable mineralogy are considered to be of cumulate type — enriched in phenocrysts of plagioclase, olivine and pyroxene.

An ankaramitic flow, with local variations to olivine and/or plagioclase--rich compositions, occurs at 430 m up the succession at the western end of the Tálknafjördur—Arnarfjördur peninsula; in the vicinity of its probable feeder dyke, it displays remarkable flow banding. Cumulate plagioclase is far more abundant as a characteristic phenocrystal phase. In the feldsparphyric flows, the crystals of plagioclase are small (2—3 mm av. diameter) and usually

aggregated into stellate groupings; in the cumulate flows, each phenocryst is isolated and idiomorphic. Many flows are characterized by cumulate plagioclase of 5—10 mm av. diameter; such flows are so numerous that correlation between measured vertical sections is difficult and individual examples are not shown in Figs. 1 and 2. Two flows are so crowded with large cumulate feldspars, 1—3 cm diameter that they resemble gabbro in their weathered outcrops. Gabbro xenoliths are not uncommon and are composed of the same idiomorphic plagioclase cemented by interstitial augite. The lowest of these flows occurs on the Patreksfjördur—Tálknafjördur peninsula, at 1000 m up the succession; its feeder dyke is exposed in the corrie behind the old whaling station at Sudureyri. Either a shallow origin for the parental magma or the development of a high-level magma reservoir seems a necessary condition for the crystallization of these cumulate phases.

Interbasaltic beds are of common occurrence. Wind-blow dusts, of red oxidized basaltic glass and crystal fragments, rarely more than 0.5 m thick have filled in the fissured surface of most flows. Occasionally, the dusts are yellow and of rhyolitic composition. Hyaloclastite deposits, with or without pillow lavas, must mark the sites of ponded surface water, and the subsequent drainage of such ponds has often reworked the palagonite deposit into stream sediments. Lignite, leaf beds and ironstones are associated with these sediments and are most abundant in the lower half of the succession along the southern shore of Arnarfjördur. Their repeated development in this area may mark the site of a Miocene valley along Arnarfjördur. It may also be significant that both the ankaramite basalt and the lower cumulate plagioclase flow peter out east of their respective feeder dykes (see below) — a feature which may be explained by an original westerly tilt to the plateau.

One hyaloclastite/pillow lava/lignite horizon at 1400 m up the succession can be traced across the whole region. Many, but not all, of the known lignite localities occur here. The true origin of the deposit is best seen in the valleys at the SE end of Arnarfjördur. Widespread flooding (by water) of a well wooded plateau must have followed a general cessation of volcanic activity. Tree moulds, in situ or transported, are common throughout the succession; one example preserved some seventy annular growth rings. Remnants of tephra cones, very similar to those of Kerid (Jakobsson, 1966) are left between the basalts in the cliffs of Selárdalur and Austmannsdalur. Thick deposits (up to 80 m) of red scoria, 3—5 cm diameter, contain thin flows of basalt and numerous 'pancake' bombs, which define dips of 30°. Cappings of hyaloclastite material are often present.

At 650 m up the succession there is a thick development of 'cone' (shield volcano) basalts. These are restricted to the Arnarfjördur area and can be examined in precipitous cliff sections, which are unlike the terraced cliffs of the plateau basalts because the thin (0.2—3 m) highly vesicular flows of non-porphyritic or feldsparphyric basalt show no scoriaceous tops or interbasaltic beds. Pahoehoe toes and ropy surfaces mark flow tops, and the picture is one of almost continuous, uninterrupted outpouring of lava. A

central, 'shield' volcano occurs at this level in the basalt plateau on the north side of Arnarfjördur, and the 'cone' basalts are most probably a remnant of its southern flank. In cliff sections to the north and south of the fjord, small unconformities can be observed between the upper surface of the cone and the overlapping and inundating plateau basalts.

The plateau basalts dip at 2—5°, the downdip direction being 160° east of true north. This dip varies locally, the lower dips of 2—3° being found in the NW part of the area, and higher dips of 5° in the SE part.

Dykes are a common feature of the plateau. Two can be identified as feeders; the connection between dyke and flow is apparent in cliff section and the flow at its source shows a well developed flow-banding near its base — a feature rarely seen in the plateau basalts. Other dykes show chilled palagonitic margins and their present outcrops must have been close to the plateau surface. These dykes trend NW—SE and in the Ósafjördur area they are closely associated with small normal faults and an intense zeolitization of the adjacent basalts. Zeolitization, not so intense, reoccurs in the western parts of the region. A few dykes trend ESE and may have their origins in later Tertiary or Pleistocene volcanism much further east. The thermal plumes rising from these may account for the linear arrangement of warm (ca. 40°C) springs from Tálknafjördur to Reykjarfjördur.

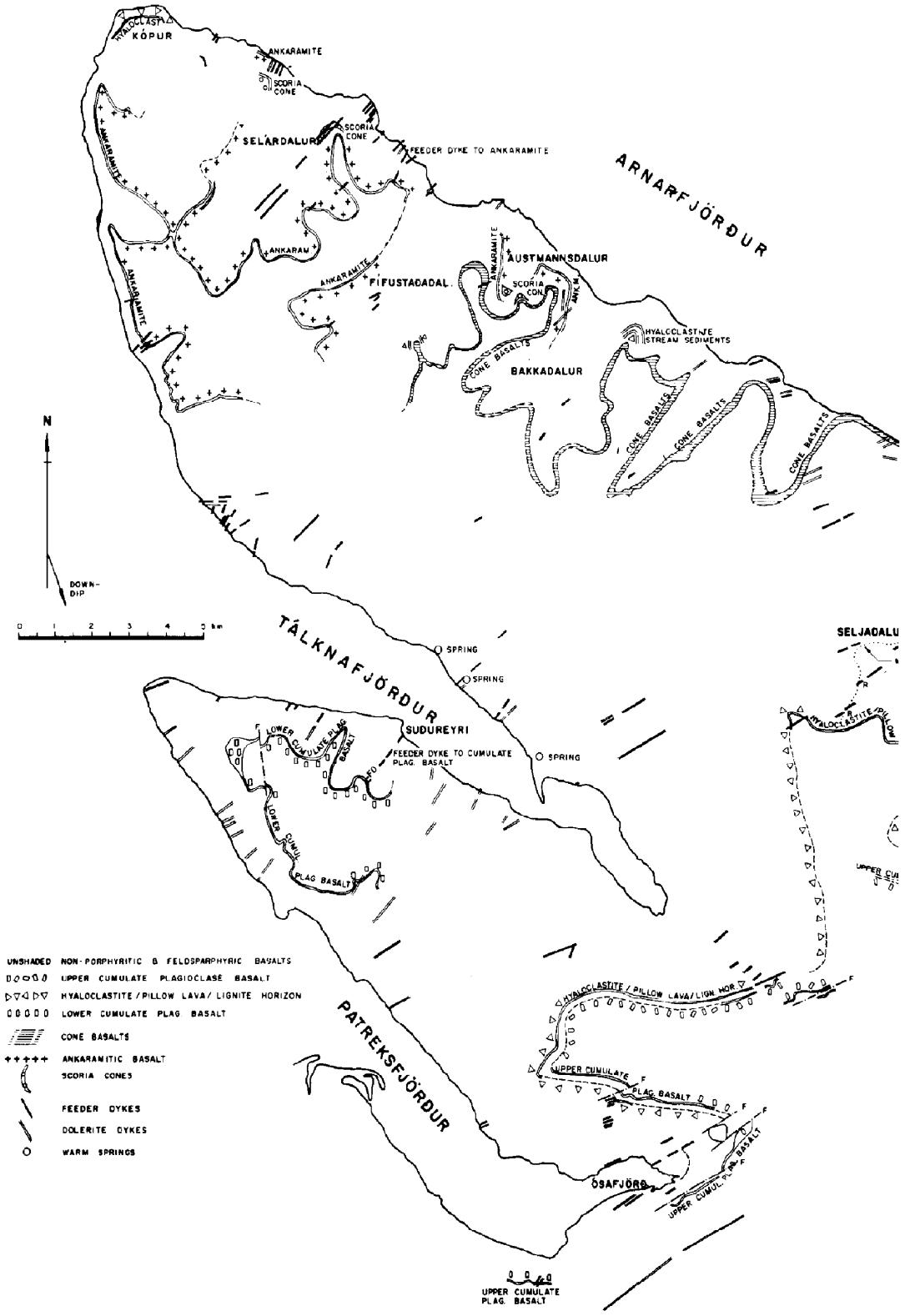
PALAEOMAGNETIC SAMPLING AND MEASUREMENTS

A sequence of lava flows, numbered AB 1 through 43, was sampled in Seljadalur valley (Fig. 1), starting at the base of continuous exposures at 60 m above sea level. The flows are separated by red sedimentary beds or scoriaceous tops. The flow above AB 30 was numbered AB 30A in the field, but palaeomagnetic results indicate it is a separate flow. The topmost two flows are porphyritic and underlie hyaloclastites at about 480 m altitude.

Three or four one-inch diameter cores were drilled from each flow and oriented by geographic or sun bearings (accuracy $\simeq 3°$). The natural remanence (NRM) of these samples (one specimen per sample) was measured with a PAR spinner magnetometer. On the basis of detailed alternating field (AF) demagnetization results on five pilot samples, a field of 200 Oe (peak) was chosen for the treatment of the remainder.

After 200-Oe treatment, it was found that the within-flow α_{95} of the palaeomagnetic field direction was always less than 10°, except in flows AB 2 and AB 34, which had been sampled in the vicinity of dykes. These have therefore been omitted from the subsequent analysis of remanence directions.

All the flows sampled have a stable normal component of magnetization, except the top two flows which are reversely magnetized. The stable component is definitely primary in origin, since the within-flow directional dispersion was much less than that between flows. Only minor zeolitization has taken place in the sequence, the main zeolites being chabazite and thomsonite. The carrier of remanence is oxidized titanomagnetite; strong-field thermomagnetic


HYALOCLASTITE
KÓPUR
ANKARAMITE
SCORIA CONE
SELÁRDALUR
FEEDER DYKE TO ANKARAMITE
ARNARFJÖRÐUR
ANKARAMITE
FÍFUSTAÐADAL
AUSTMANNSDALUR
HYALOCLASTITE STREAM SEDIMENTS
CONE BASALTS
BAKKADALUR
N
DOWN-DIP
0 1 2 3 4 5 km
TÁLKNAFJÖRÐUR
SPRING
SUÐUREYRI
LOWER CUMULATE PLAG. BASALT
FEEDER DYKE TO CUMULATE PLAG. BASALT
SELJADALU
HYALOCLASTITE / PILLOW
UNSHADED NON-PORPHYRITIC & FELDSPARPHYRIC BASALTS
UPPER CUMULATE PLAGIOCLASE BASALT
HYALOCLASTITE / PILLOW LAVA / LIGNITE HORIZON
LOWER CUMULATE PLAG. BASALT
CONE BASALTS
ANKARAMITIC BASALT
SCORIA CONES
FEEDER DYKES
DOLERITE DYKES
WARM SPRINGS
PATREKSFJÖRÐUR
HYALOCLASTITE / PILLOW LAVA / LIGN. HOR.
UPPER CUMULATE PLAG. BASALT
ÖSAFJÖRÐ
UPPER CUMULATE PLAG. BASALT

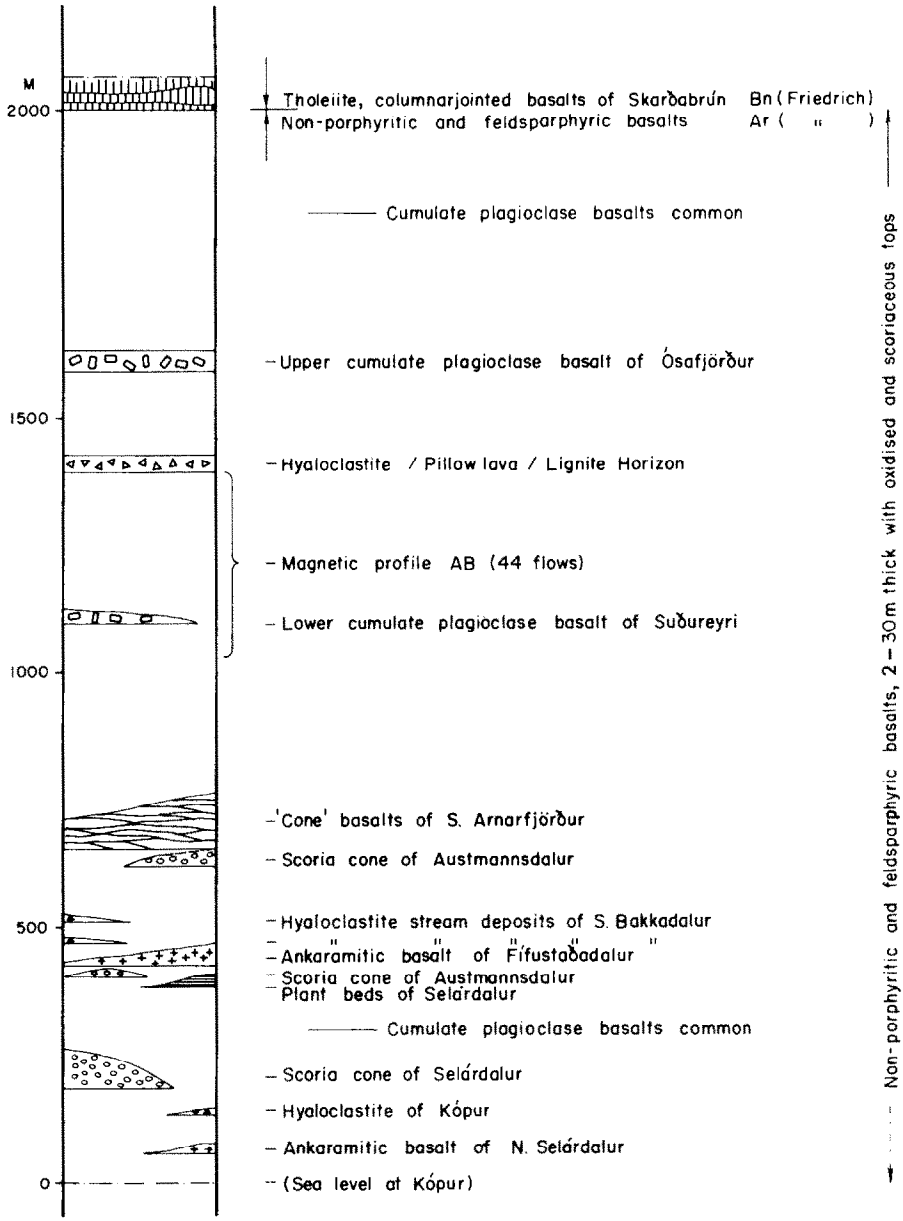


Fig. 2. Generalized succession of basalt lava flows and interbasaltic beds for the Patreksfjördur—Arnarfjördur region.



Fig. 1. Geological map of the Patreksfjördur—Arnarfjördur region, NW-Iceland, showing distinctive horizons, dykes, faults and other features (after J. Preston). Inset shows location of this region ("Pr"), area "Fr" mapped by Friedrich (1966) and palaeomagnetic sampling area "Kr" of Kristjánsson (1968). Also shown is an aeromagnetic profile from Serson et al. (1968) and sand dredge localities *A* through *R* referred to in the text.

TABLE I

Mean magnetic properties of lava flows from profile *AB*

Flow no.	No. of samples measured	D	I	α_{95}	λ	ϕ	$\bar{J}_0$	$\bar{k}$
1	3	14.2	59.2	3.3	63.5	131.5	1.1	0.35
2	4	normal		>10°			2.5	2.65
3	4	17.9	61.0	8.2	64.3	124.4	0.9	1.8
4	3	306.0	72.3	5.9	63.9	254.2	4.1	2.15
5	3	351.7	61.1	2.8	66.0	171.5	2.3	1.1
6	3	78.6	89.0	1.8	66.0	341.1	2.2	1.65
7	3	56.9	80.8	5.4	69.4	23.5	1.9	1.4
8	3	27.6	79.7	3.0	78.8	31.1	3.4	1.2
9	3	57.9	78.1	6.0	67.6	36.1	5.3	1.6
10	3	52.3	77.7	6.7	69.4	40.2	5.5	1.1
11	3	36.2	73.3	3.1	72.2	70.9	12.3	2.4
12	3	17.0	72.1	4.2	78.3	104.8	5.1	1.55
13	4	30.4	71.0	3.3	72.2	86.5	4.2	0.55
14	3	11.5	65.5	6.6	71.0	131.9	5.2	1.45
15	3	305.8	83.1	8.7	70.4	301.7	6.2	1.4
16	3	328.4	65.6	2.8	65.6	214.7	7.2	1.55
17	3	316.2	39.7	6.4	38.6	211.1	4.5	1.3
18	3	359.2	75.4	3.8	86.8	163.1	5.7	1.85
19	3	33.6	72.7	2.3	72.7	76.2	4.0	1.15
20	3	356.8	66.0	3.3	72.5	163.4	2.1	1.1
21	3	34.3	69.1	8.3	68.5	86.9	4.7	1.45
22	3	334.8	81.3	2.4	78.6	297.3	6.6	1.3
23	3	309.1	82.3	1.3	71.3	297.2	19.0	1.8
24	3	326.1	82.1	3.7	75.8	299.0	7.2	1.7
25	3	27.6	84.7	4.1	74.2	354.5	5.5	1.0
26	3	17.2	58.9	4.4	62.2	127.0	3.1	1.05
27	3	341.0	80.3	1.0	81.1	293.0	6.6	1.3
28	3	17.4	75.4	1.8	81.8	80.9	3.5	1.05
29	4	266.4	58.5	6.3	33.8	267.9	1.0	1.8
30	3	305.9	48.6	2.5	41.2	225.9	0.6	1.5
30A	3	312.7	55.9	9.5	50.0	223.1	1.2	1.4
31	3	354.1	84.0	1.8	77.4	330.8	2.9	1.05
32	3	39.9	82.0	6.4	74.2	14.8	2.8	1.6
33	4	58.6	81.0	4.4	69.0	22.2	2.6	1.45
34	4	normal		>10			4.2	2.25
35	4	57.9	77.2	7.3	66.9	39.8	2.3	1.2
36	3	34.9	78.1	3.4	76.1	44.3	4.7	1.9
37	3	54.0	80.7	7.3	70.2	24.5	2.9	1.2
38	3	63.4	77.7	3.4	65.3	35.3	2.1	0.9
39	3	346.3	75.1	3.0	82.9	221.2	3.3	0.9
40	3	296.3	84.3	2.2	68.3	308.1	3.1	0.75
41	3	59.6	69.3	6.5	58.5	60.7	3.5	1.5
42	3	213.5	−64.3	6.1	63.5	97.2	3.1	1.1
43	3	217.6	−64.6	2.4	62.3	91.5	2.7	0.8

D = declination and I = inclination of mean field direction after tilt correction. α_{95} = 95% confidence angle for mean field direction, in degrees. $\bar{J}_0$ = logarithmic average NRM intensity, in 10^{-3}G. $\bar{k}$ = logarithmic average initial susceptibility, in 10^{-3}G/Oe. λ = latitude of northern virtual geomagnetic dipole. ϕ = longitude of northern virtual geomagnetic dipole, east of Greenwich.

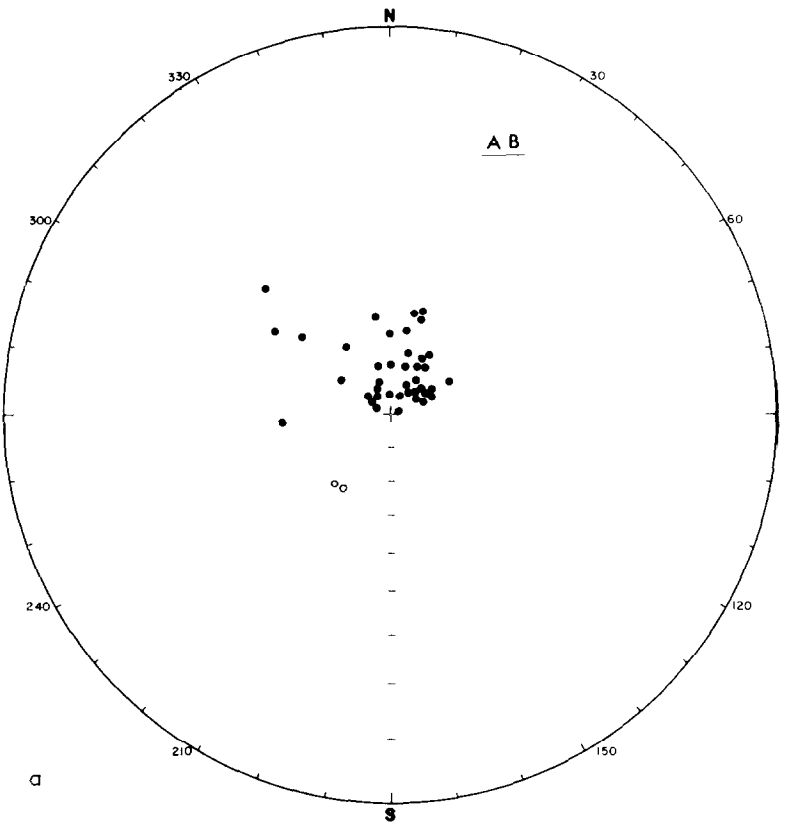
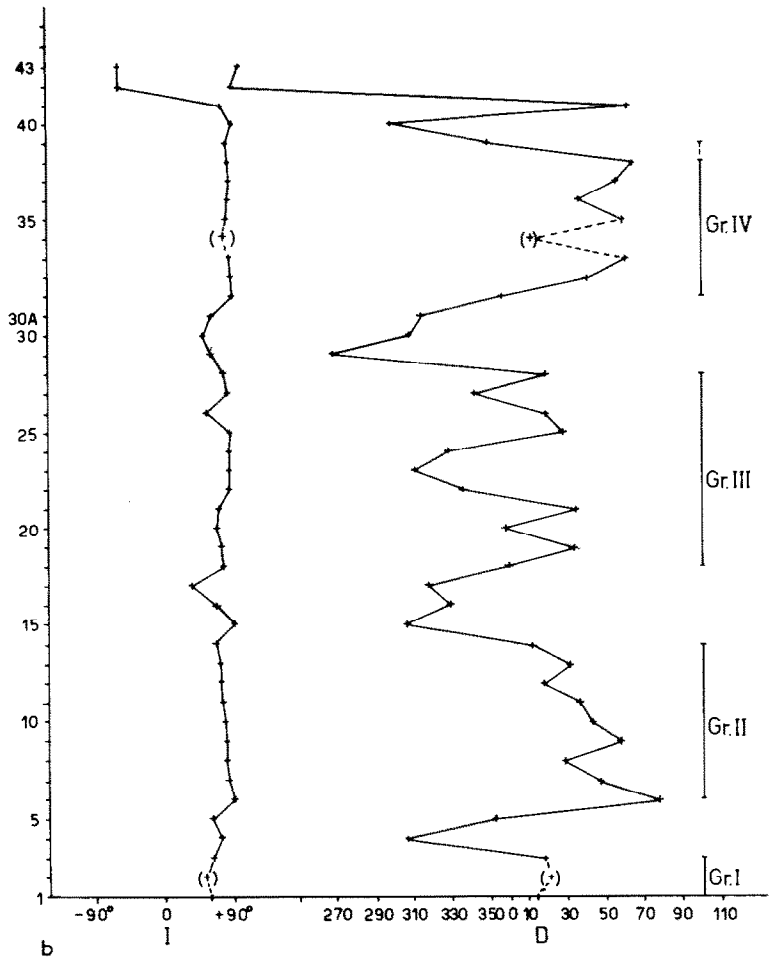


Fig. 3. Flow-average palaeomagnetic field directions, after 200-Oe treatment and tilt correction: (a) on a stereographic projection (Wulff net); (b) as a function of position in the profile.

curves from four lava specimens show Curie points $\geq 550°C$, but a fifth, as well as specimens from dykes cutting AB 2 and AB 38, had lower Curie points.

Directional results from the individual flows, corrected for regional dip, are shown in Table I and in Fig. 3a. Overall results from the normally magnetized flows only are given below, and compared to data obtained by Kristjánsson (1968) farther north:

	N	$\overline{D}$	$\overline{I}$	R	k	δ	α_{95}	ϕ	λ
Sequence AB	40	0.1	76.8	38.57	27.3	15.3	4.4	88.0	113
Kristjánsson (1968)	60	0.5	73.0	55.42	12.9	22.5	5.3	82.5	155



Here, N is the number of flows averaged, $\bar{D}$ and $\bar{I}$ the declination and inclination of the mean magnetic field, and R, k, δ and α_{95} are statistical parameters (Irving, 1964). ϕ and λ are the latitude and east longitude of the resultant northern palaeomagnetic pole (assuming a centred dipole field), with error oval $\delta m \simeq \delta p \simeq 8°$.

The higher dispersion and lower mean inclination of the 1968 results are probably due to the fact that collection was then made from several profiles straddling a short normal palaeomagnetic epoch. It is evident that meaningful reproducible estimates of long-term palaeomagnetic mean directions or dispersions can only be obtained from large numbers (> 100) of magnetically stable lava flows covering several geomagnetic epochs. Even so, mean palaeomagnetic pole positions will depend on the model of the field used in the analysis, because of the non-linear relation between palaeoinclination and virtual palaeolatitude. Thus, the mean poles derived from the above two sets

of data would be some 3° different if derived by Method II of Irving (1964, p. 70). These and various other sources of systematic errors in this kind of work, including the effect of original tilt of strata (as inferred above) tend to be overlooked by those claiming 'fits' of palaeomagnetic data to elaborate tectonic or geomagnetic reconstructions.

REMANENCE INTENSITY AND POLARITY

Histograms of NRM intensities of lava samples from the present collection are shown in Fig. 4. The overall arithmetic mean intensity from 44 flows is $4.2 \cdot 10^{-3}$ gauss, and the logarithmic mean is $3.4 \cdot 10^{-3}$ gauss. The logarithmic susceptibility mean value is $1.3 \cdot 10^{-3}$ G/Oe. These are similar to values previously published by Kristjánsson (1970) from Tertiary areas of Iceland. Results from dykes in Seljadalur (Fig. 4) are not systematically different from the lava results.

A series of normal-polarity flows as thick as the Seljadalur normal series, being at least 42 flows, is unusual for the Tertiary of Iceland and may be of wider stratigraphic importance. No parallel section has, so far, been measured but some flows or flow groups can be traced laterally for distances of 20–30 km (Fig. 1) and, with the help of other distinctive 'marker' horizons, palaeomagnetic correlations may be possible over tens of kilometres. It is interesting that the series ends at two highly porphyritic flows, but correlations of magnetic polarity and lithology, as indicated by Friedrich (1966, see above) may be coincidental.

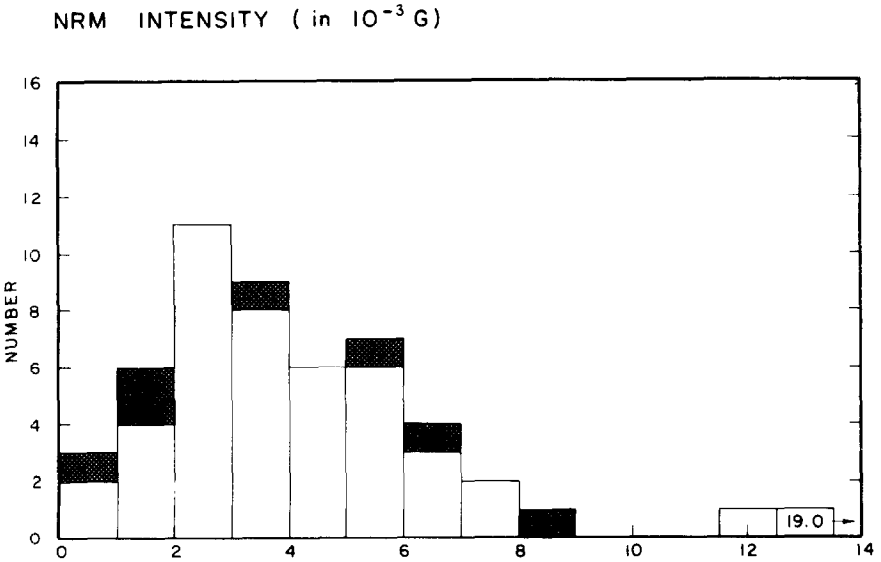


Fig. 4. Histogram of natural remanent intensity for lavas and dykes (shaded) from profile *AB*, in cgs volume units.

This normal series is likely to be a contributor to the 300 gamma positive magnetic anomaly seen over the region in the profiles of Serson et al. (1968); see inset of Fig. 1. However, further palaeomagnetic mapping is needed before this magnetic group can be correlated with anomalies in adjacent flight lines or with magnetic groups elsewhere in the country, in the manner attempted by Piper (1973). Of the local dykes sampled, five were reversely magnetized (see Fig. 1), one was normal and one unstable. These results support the views of Kristjánsson (1970) that dykes may be of minor importance as sources of aeromagnetic anomalies over Iceland.

SECULAR VARIATION

The work of e.g. Wensink (1964), Kristjánsson (1968) and Kristjánsson and Deutsch (1973), suggests that the palaeomagnetic directions of successive flows from Tertiary lava profiles in Iceland and western Greenland are generally random spot readings of the time-varying, local, geomagnetic field. Systematic grouping of such readings around a mean position, or along a path, is rarely seen and may be fortuitous except in 'flow units' or where a polarity transition or a large deviation of the field is recorded through several flows.

Wilson (1970) and Wilson et al. (1972) in their study of British Tertiary basalt stratigraphy did discover well defined groups of flows with similar palaeomagnetic directions. Wilson (1970), using a method suggested by Hospers (1954) for Icelandic lava flows, demonstrated their use in correlating neighbouring sections over a lateral distance of 3.6 km, even though exposures were incomplete.

The Seljadalur lava sequence also possibly contains groups of flows with similar palaeomagnetic directions. This can be seen in Fig. 3b and the groupings are summarized below.

Group I

Flows 1—3. An ill-defined group; the individual flows may be well separated in time.

Flows 4,5. These show considerable deviations in their palaeomagnetic directions.

Group II

Flows 6—14, 72 m thick. A well-defined coherent group of feldsparphyric basalts.

Flows 15, 16, 17. These have markedly deviating palaeomagnetic directions.

Group III

Flows 18—28, 96 m thick. A moderately well-defined and coherent sequence of feldsparphyric basalts.

Flows 29, 30, 30A. These have marked deviations in their palaeomagnetic directions.

Group IV

Flows 31—39 (41?), 77 (87) m thick. A well-defined coherent group of non-porphyritic basalts. The upper limit of the sequence is somewhat uncertain.

Group V

Flows 42, 43. Two cumulate (feldsparphyric) flows of similar, reversed, palaeomagnetic directions, overlain by hyaloclastites.

Very little data can be given, as yet, on the mineralogical or petrochemical characters of these flows and groupings. Preliminary major-element and trace-element analyses do not illustrate any group features, such as batch magma generation or progressive differentiation. Each flow appears to have similar primitive compositions.

No flow units have been recognized in this profile of 44 flows; red dust horizons and tree moulds are of common occurrence and evidence a separate cooling history for each flow. A minimum estimate for the time interval between would be about 200 years, both from the magnetic and the palaeontological evidence. The interbasaltic beds cannot be used directly, as Musset et al. (1972) have used them, to infer longer intervals. Some examples are most certainly lava crusts weathered in situ over long periods of time; many, and in particular the Seljadalur examples, are wind blown dusts of oxidized glassy tephra. Such deposits can accumulate very rapidly in the vicinity of fountaining lava; they are usually clearly defined against the underlying rock. The glass fragments would weather readily to clay or laterite.

The mean time between successive lava flows can be deduced from the east Iceland data of Dagley et al. (1967) and more recent radioactive datings as being of the order of 10,000 years. Doell and Cox (1965), however, record interflow intervals of 100—200 years within the summit calderas of the Hawaiian shield volcanoes and 10,000 years between successive overlapping flows on their flanks.

This more realistic picture of volcanic activity can be applied to the Seljadalur lava sequence. A group of active fissures across the basaltic plateau would correspond in degree of activity with the summit region of a mature shield volcano and rapidly build up an elongated ridge to a height of 70—100 m. This would produce a flow group with interflow intervals of approximately 200 years. Limiting 'hydrostatic' pressure on the rising magma column would result in a lateral migration — some tens of kilometres — of the active fissure zone. A similar group of flows, lens-shaped in cross-section, would flood the lower ground and only occasionally would lava overlap marginally onto the earlier extrusives. A considerable interval — the inter group interval — of some tens of thousands of years duration would then elapse before an overlapping group buried this new plateau level.

SUSCEPTIBILITY

It is interesting to compare the logarithmic average susceptibility of the lava profile to the susceptibility of rock in sand samples dredged off shore. Since in large collections the mean susceptibility probably is mostly a function of magnetite content (Kristjánsson and Deutsch, 1973) such comparisons will yield an estimate of the relative enrichment in heavy minerals which the rock has experienced during weathering and transportation.

Seventeen samples of dredged sand, kindly provided by Dr. K. Thors of the Marine Research Institute, were investigated. After elimination of carbonates, which made up a varying fraction (3—95%) of the samples, the following mean susceptibility values by weight were found:

Sample group	Locations in inset of Fig. 1	Susceptibility 10^{-3} emu/g
Arnarfjördur	*CDEFG*	1.1
Other Vestfirdir	*BHIK*	0.55
Faxaflói and Breidifjördur	*AJQR*	0.42
Reykjanes peninsula	*LMNP*	0.28
Lava profile AB Kristjánsson (1968)	(assuming density of 2.7 g/cm^3)	0.48

It is seen that the susceptibility of the sand is not greatly higher than that of the country rock. The low values in the dredge samples off Reykjanes probably reflect the low susceptibility of rocks in the palagonite formation (Kristjánsson, 1970) and are similar to values from the glacial outwash sands of Southern Iceland.

The mechanism of magnetite concentration, tentatively suggested by Einarsson (1970) to account for certain magnetic anomalies on the shelf around Iceland, does not therefore seem to be generally of importance. However, on the beach at Tjaldanes in Arnarfjördur, below gabbro outcrops belonging to the above-mentioned central volcano, wave action is indeed able to separate and concentrate titanomagnetite grains to a remarkable degree: sand samples collected here have shown susceptibility values of up to $7.5 \cdot 10^{-3}$ weight units, i.e. 15 times the average from sequence *AB*. One sand sample has been chemically analyzed (P. Imsland, personal communication, 1973) after hand-picking with a strong magnet and contained 61% total iron as Fe_2O_3.

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